

Closed-Form Evaluation of $\sin 18^\circ$ and $\cos 18^\circ$

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Solution

Let

$$A = 18^\circ,$$

and write

$$s = \sin A, \quad c = \cos A.$$

Since

$$5 \times 18^\circ = 90^\circ,$$

we have

$$\cos 5A = \cos 90^\circ = 0.$$

Now,

$$5A = 2A + 3A,$$

so the angle addition formula gives

$$\cos 5A = \cos(2A + 3A) = \cos 2A \cos 3A - \sin 2A \sin 3A.$$

Therefore,

$$\cos 2A \cos 3A = \sin 2A \sin 3A.$$

Using the multiple-angle formulae,

$$\cos 2A = 2c^2 - 1,$$

$$\cos 3A = 4c^3 - 3c,$$

$$\sin 2A = 2sc,$$

and

$$\sin 3A = 3s - 4s^3,$$

we obtain

$$(2c^2 - 1)(4c^3 - 3c) = (2sc)(3s - 4s^3).$$

Using

$$s^2 + c^2 = 1,$$

we have

$$s^2 = 1 - c^2.$$

Hence,

$$\begin{aligned} (2c^2 - 1)(4c^3 - 3c) &= 2sc(3s - 4s^3) \\ &= 2cs^2(3 - 4s^2) \\ &= 2c(1 - c^2) [3 - 4(1 - c^2)] \\ &= 2c(1 - c^2)(4c^2 - 1). \end{aligned}$$

Therefore,

$$(2c^2 - 1)(4c^3 - 3c) = 2c(1 - c^2)(4c^2 - 1).$$

Since $c = \cos 18^\circ \neq 0$, we may divide by c to obtain

$$(2c^2 - 1)(4c^2 - 3) = 2(1 - c^2)(4c^2 - 1).$$

Expanding both sides gives

$$8c^4 - 10c^2 + 3 = -8c^4 + 10c^2 - 2.$$

Hence,

$$16c^4 - 20c^2 + 5 = 0.$$

Let

$$x = c^2.$$

Then

$$16x^2 - 20x + 5 = 0.$$

Applying the quadratic formula,

$$\begin{aligned} x &= \frac{20 \pm \sqrt{400 - 320}}{32} \\ &= \frac{20 \pm \sqrt{80}}{32} \\ &= \frac{20 \pm 4\sqrt{5}}{32} \\ &= \frac{5 \pm \sqrt{5}}{8}. \end{aligned}$$

Since 18° is acute and $\cos 18^\circ$ is the larger positive possibility,

$$\cos^2 18^\circ = \frac{5 + \sqrt{5}}{8}.$$

Thus,

$$\cos 18^\circ = \sqrt{\frac{5 + \sqrt{5}}{8}}.$$

Putting the expression into standard radical form,

$$\boxed{\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}}.$$

We now use

$$\sin^2 A + \cos^2 A = 1.$$

Therefore,

$$\begin{aligned} \sin^2 18^\circ &= 1 - \cos^2 18^\circ \\ &= 1 - \frac{5 + \sqrt{5}}{8} \\ &= \frac{3 - \sqrt{5}}{8}. \end{aligned}$$

Hence,

$$\sin 18^\circ = \sqrt{\frac{3 - \sqrt{5}}{8}}.$$

To express this in standard form, observe that

$$\begin{aligned}\left(\frac{\sqrt{5}-1}{4}\right)^2 &= \frac{(\sqrt{5}-1)^2}{16} \\ &= \frac{6-2\sqrt{5}}{16} \\ &= \frac{3-\sqrt{5}}{8}.\end{aligned}$$

Since $\sin 18^\circ > 0$,

$$\boxed{\sin 18^\circ = \frac{\sqrt{5}-1}{4}}.$$

Therefore, the required closed forms are

$$\boxed{\sin 18^\circ = \frac{\sqrt{5}-1}{4}}$$

and

$$\boxed{\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}}.$$

The End